

ARUN ARUNACHALAM

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PROFESSIONAL SUMMARY

Data and AI technology executive with 17+ years architecting enterprise data platforms, AI-enabled products, and data governance frameworks across capital markets at Bank of America. Deep expertise in building and scaling data engineering pipelines, GenAI platforms (RAG, agentic AI), and analytics infrastructure that drive intelligent decision-making across complex global organizations. Proven record of ensuring data quality, integrity, and security while modernizing legacy systems into cloud-native, event-driven architectures that generate measurable business outcomes. Combines executive stakeholder leadership with hands-on command of distributed data systems, ML-based intelligence, and AI-assisted delivery to accelerate an enterprise's data and analytics journey.

SIGNATURE IMPACT

- **\$8M** enterprise transformation → **40% performance improvement, 30% cost reduction**
- **AI-assisted engineering** → **35% faster delivery, 45% productivity gain**
- Scaled platforms from 10 to **50+ systems** with **99.97% uptime**
- Enabled **~\$2B capital efficiency** via risk platform transformation

CORE CAPABILITIES

AI & GenAI Platforms (RAG, Agentic AI) | Data Engineering (Spark, Databricks, ETL/ELT) | Data Governance, Quality & Integrity | Cloud-Native Architecture (AWS, Kubernetes) | Event-Driven Systems (Kafka) | Enterprise Data Strategy | Trading & Risk Platforms | Regulatory & Compliance Systems (FRB, FINRA, SEC) | Executive Stakeholder Leadership | Global Team Leadership | Platform Modernization

PROFESSIONAL EXPERIENCE

Bank of America — Director, Global Risk & Capital Markets Technology (2015 - Present)

- Drove enterprise data strategy across trading and data systems supporting **\$1T+ daily processing**, ensuring data quality, integrity, and security at scale
- Architected and delivered cloud-native, event-driven data platforms (AWS, Kafka, Kubernetes) enabling real-time analytics and intelligent decision-making across capital markets
- Built and governed data engineering pipelines (Spark, Databricks, ETL/ELT) powering risk, pricing, and trading analytics across global business lines
- Implemented GenAI platforms including RAG pipelines and agentic AI systems to accelerate delivery and embed AI-driven intelligence into operational workflows
- Applied ML-based anomaly detection to improve data quality monitoring and operational intelligence across mission-critical systems
- Delivered real-time trading, pricing, and risk platforms improving throughput by **40%**, underpinned by high-availability data infrastructure at **99.97% uptime**
- Ensured regulatory compliance (FRB, FINRA, SEC, SOX) across enterprise data and analytics systems, aligning data governance with firm-wide risk and audit requirements

- Led AI-assisted SDLC workflows driving **35% faster delivery** and **45% productivity gain** across engineering teams
- Built and led global engineering teams (**30+ across US/UK/India**) driving **15% velocity improvement** across data and platform programs

Bank of America — VP, Structured Products & Mortgage Technology (2011 - 2015)

- Led consolidation and governance of risk data systems contributing to **~\$2B capital efficiency**
- Built real-time pricing and analytics platforms for structured products and mortgage trading, delivering high-integrity data to front-office decision makers
- Designed data pipelines and analytics infrastructure supporting complex financial instrument valuation and risk reporting

Bank of America — AVP, Whole Loan Trading Desk Support (prior to 2011)

- Architected and delivered end-to-end Java EE trading support applications with integrated data flows for whole loan trading operations
- Introduced CI/CD practices reducing release cycle from monthly to weekly, improving data product delivery cadence and quality

Senior Programmer Analyst — Knight Trading (2004)

DB2 / WebSphere Administrator & Sr. Java Developer — Kyocera Mita (2003)

Sr. Java Web Developer — Mellon Investor Services (2001)

Sr. Java GUI Developer — Thomson Financials (2001)

Sr. Java GUI Developer — Icor Brokerage (2000)

AI & INNOVATION

- Building GenAI platforms (RAG pipelines, agentic AI systems) designed for enterprise-scale, real-world data and analytics applications
- Applied ML for anomaly detection and operational intelligence, improving data quality and system reliability across mission-critical platforms
- Driving AI-assisted SDLC workflows that improve delivery speed, quality, and engineering productivity across data product development

EDUCATION

Bachelor of Technology — Anna University

SKILLS

Leadership: Executive stakeholder management, enterprise data strategy, global team leadership (30+ engineers), P&L ownership, strategic planning, program management, SDLC, Agile, ITIL, SOX, FRB/FINRA/SEC compliance **Data & AI:** GenAI (RAG pipelines, agentic AI), ML anomaly detection, AI-assisted SDLC, data governance, data quality & integrity, Spark, Databricks, ETL/ELT, Snowflake, Hadoop, Cassandra, MongoDB **Architecture:** Cloud-native architecture (AWS, Kubernetes), enterprise architecture, event-driven systems (Kafka), distributed systems, high-availability platforms (99.97% uptime) **Engineering:** Java, Python, Node.js, SQL, Oracle, Sybase, REST/SOA, WebLogic, Kafka, Kubernetes, AWS **Tools:** ServiceNow, JIRA, Git, Visio, Microsoft Project